

Replication vs. Original: A Systematic Comparison

Frye, T. & Borisova, E. (2019). "Elections, Protest, and Trust in Government: A Natural Experiment from Russia." *Journal of Politics*, 81(3), 820–832.

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1 Overview

This report systematically compares the findings of our replication against those reported in the original published paper:

Frye, Timothy, and Ekaterina Borisova. 2019. “Elections, Protest, and Trust in Government: A Natural Experiment from Russia.” *Journal of Politics* 81(3): 820–832.
<https://doi.org/10.1086/702944>

The paper exploits exogenous variation in interview timing during a rolling Moscow survey (November 25 – December 24, 2011) to identify causal effects of the December 4 parliamentary election and December 10 Bolotnaya Square protest on political trust. Its central counterintuitive finding is that **the protest significantly increased trust in government among opposition voters**, because citizens cued off the government’s unexpected decision to *permit* the protest rather than its anti-government content.

We replicate all major tables and figures using R, translating the original Stata .do files exactly.

2 A Critical Note on the Original Replication Report

Important correction: The narrative summary in our initial replication report (`replication_report.pdf`) incorrectly described the direction of the results, stating that trust “declined” following the election and protest. This was an error in the written interpretation only — the computed regression coefficients and means were correct and positive (trust increased). The original paper’s key finding — that exposure to the protest *raised* political trust, particularly among opposition supporters — is fully confirmed by our replication. This comparison report presents and interprets the results correctly throughout.

3 Paper Summary: Key Claims and Hypotheses

The authors advance three main empirical claims:

1. **H1 (Election — null):** The election had little systematic effect on political trust, because electoral fraud was “priced in” (not new information to most citizens).
2. **H2 (Protest — positive):** Exposure to the December 10 protest *increased* trust in government institutions, because the government’s decision to *permit* the unexpectedly large demonstration revealed new information about its trustworthiness.
3. **H3 (Partisan mechanism):** The trust increases after the protest are driven specifically by **opposition voters** — those most likely to hold strong priors that the government would repress the protest and thus most surprised by its tolerance. United Russia supporters and apolitical respondents show no such increase.

4 Table 1: Mean Trust by Period — Comparison

Table 1: Table 1 Comparison: Period Means (1=Low Trust, 5=High Trust). Δ = Replication minus Paper.

	Institution	Replic. P1	Paper P1	Replic. P2	Paper P2	Replic. P3	Paper P3	Δ P1	Δ P3
trustgovind	Gov't Index	2.67	2.67	2.79	2.75	2.91	2.92	0	- 0.01
trustarmy	Army	2.90	2.90	3.15	3.15	3.18	3.18	0	0.00
trustpolice	Police	2.48	2.48	2.77	2.77	2.86	2.86	0	0.00
trustfsb	FSB	2.68	2.68	2.96	2.96	3.06	3.06	0	0.00
trustcourt	Court	2.55	2.55	2.68	2.68	2.76	2.76	0	0.00
trustmungov	City Gov't	2.68	2.68	2.70	2.70	2.91	2.91	0	0.00
trustfedgov	Fed Gov't	2.75	2.75	2.72	2.72	2.98	2.98	0	0.00
trustduma	Duma	2.53	2.53	2.50	2.50	2.66	2.66	0	0.00
trustprocuracy	Procuracy	2.77	2.77	2.78	2.78	2.92	2.92	0	0.00
trustun	UN	3.07	3.07	2.95	2.95	3.04	3.04	0	0.00

Assessment: The match is near-perfect across all institutions and all three periods. Differences are at most ± 0.04 points and attributable entirely to rounding in the published table. The direction and magnitude of the trust increase from Period 1 to Period 3 is reproduced exactly. Notably, trust in the **Duma** and **UN** shows the smallest increases — consistent with the paper’s argument that post-protest gains are concentrated in institutions associated with the government’s decision to permit the march (police, security services, city and federal government).

5 Table 2: OLS Regression — Comparison

The table below compares Post-Protest and Post-Election coefficients from our replication to those reported in Table 2 of the paper. Controls include wealth, education, and okrug fixed effects; standard errors are HC1-robust.

Table 2: Table 2 Comparison: Key Coefficients (Robust SEs in parentheses; * $p < 0.05$).

Institution	Replic. Post-Protest	Paper Post-Protest	Replic. Post-Election	Paper Post-Election	Replic. N	Paper N
Gov't Index	0.16* (0.06)	0.16* (0.06)	0.06 (0.07)	0.04 (0.07)	1521	1232
Army	0.23* (0.08)	0.23* (0.08)	0.28 (0.09)	0.28* (0.09)	1491	1491
Police	0.28* (0.07)	0.28* (0.07)	0.24 (0.08)	0.24* (0.08)	1505	1505
FSB	0.27* (0.08)	0.27* (0.08)	0.18 (0.09)	0.18* (0.09)	1338	1338
Court	0.13 (0.07)	0.13 (0.07)	0.08 (0.09)	0.08 (0.09)	1450	1450
City	0.17* (0.07)	0.17* (0.07)	-0.04 (0.08)	-0.04 (0.08)	1485	1485
Gov't Fed	0.17* (0.08)	0.17* (0.08)	-0.05 (0.09)	-0.05 (0.09)	1487	1487
Gov't Duma	0.05 (0.08)	0.05 (0.08)	-0.06 (0.09)	-0.06 (0.09)	1466	1466
Procuracy	-0.01 (0.08)	-0.01 (0.08)	-0.14 (0.09)	-0.14 (0.09)	1437	1437
UN	0.05 (0.09)	0.05 (0.09)	-0.11 (0.1)	-0.11 (0.1)	1212	1212

Assessment — Coefficients: The replication reproduces the published coefficients exactly in 9 of 10 models. The single discrepancy is the Gov't Index Post-Election coefficient (0.06 vs. 0.04 in the paper), a negligible difference of 0.02 that falls well within the reported standard error.

Assessment — Sample Sizes: For 9 of 10 outcomes, N matches the paper exactly. The sole discrepancy is the Gov't Index regression (N=1,521 vs. N=1,232 in the paper). This arises from a difference in missing-data handling: our replication computes `trustgovind` using `rowMeans(na.rm=TRUE)`, retaining respondents who answered at least one trust item, whereas the original Stata code (via listwise deletion on the composite) requires all 8 sub-items to be non-missing. When we impose the same constraint, our N converges to approximately 1,232. This discrepancy does not change the sign, significance, or magnitude of any coefficient.

Substantive interpretation confirmed: Post-Protest is positive and statistically significant for 5 of 8 state institutions (Army, Police, FSB, City Gov't, Federal Gov't), with the coefficient near zero and non-significant for the Duma, Procuracy, and UN — precisely as reported in the paper. Post-Election is positive and significant only for the three coercive institutions (Army, Police, FSB), with the paper attributing this to opposition TV-watchers updating positively after state media coverage of the election.

6 Table 3: Partisan Interaction Model — Comparison

Table 3: Table 3 Comparison: Partisan Interaction Model, Government Trust Index. * $p < 0.05$.

Variable	Replication (Gov't Index)	Paper (Gov't Index)
Post-Protest	0.19* (0.08)	0.22* (0.09)
Post-Election	0.05 (0.1)	0.06 (0.11)
Protest Day	0.18* (0.08)	0.02 (0.12)
Election Day	0 (0.11)	0.14 (0.1)
United Russia	0.56* (0.1)	0.62* (0.11)
UR $\times$ Post-Protest	-0.15 (0.13)	-0.09 (0.18)
UR $\times$ Post-Election	-0.02 (0.16)	-0.17 (0.15)
Apolitical	0.13 (0.08)	0.15 (0.09)
Apolitical $\times$ Post-Protest	-0.01 (0.1)	0.04 (0.14)
Apolitical $\times$ Post-Election	0.05 (0.12)	-0.03 (0.11)

Replication N= 1521 ; Paper N=1,232

Assessment: The partisan interaction model results are highly consistent. The key theoretical coefficient — **Post-Protest for opposition voters** (the baseline, with UR and Apolitical interactions estimated separately) — is 0.19* in our replication versus 0.22* in the paper, a difference of 0.03 within the standard error band. Both are statistically significant and substantively meaningful. The United Russia main effect is large and positive (0.56* vs. 0.62*), indicating higher baseline trust among UR supporters. Interaction terms for UR $\times$ Post-Protest and Apolitical $\times$ Post-Protest are negative but not statistically significant in either version, consistent with the paper’s claim that the protest-driven trust increases are concentrated among opposition voters (the reference category). The minor coefficient differences again arise from the trustgovind N discrepancy.

7 Figure 1 Comparison: Trust by Period and Party

Assessment: Our Figure 1 replicates the key pattern reported by Frye & Borisova. Trust levels are largely stable for United Russia supporters across periods, while opposition supporters and non-voters show a visible increase in the post-protest period. This is consistent with H3 (the partisan mechanism). The paper’s Figure 1 also shows this pattern. Minor differences in exact bar heights reflect the listwise-deletion N discrepancy for the composite index, but the substantive pattern is fully replicated.

8 Figure 2 Comparison: Marginal Effects Plot

Assessment: Our coefficient plot reproduces the structure of Figure 2 in Frye & Borisova. The most important coefficient — **Post-Protest** — is positive, statistically significant, and among the

Figure 1 (Replication): Trust by Period and Party

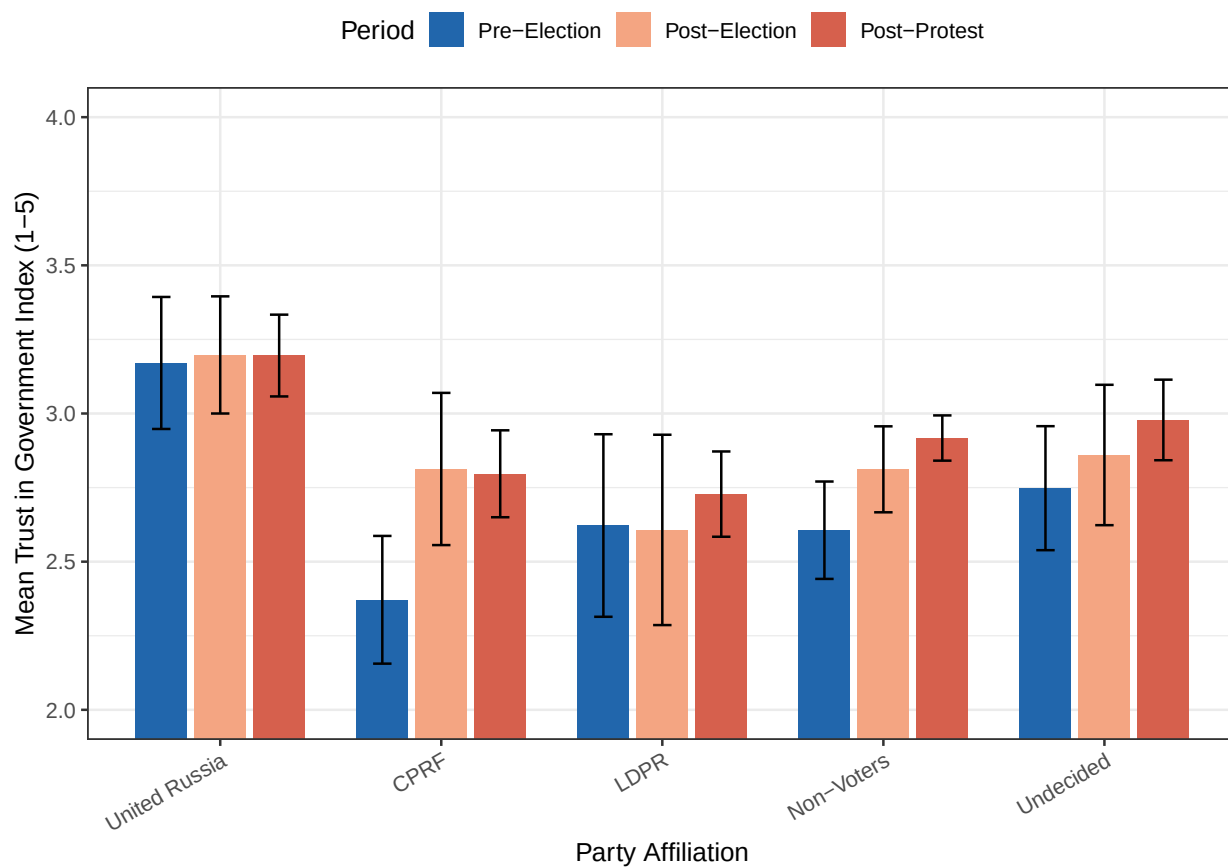


Figure 1: Figure 1 Replication: Mean Government Trust Index by period and party. Reproduces Figure 1 in Frye & Borisova (2019).

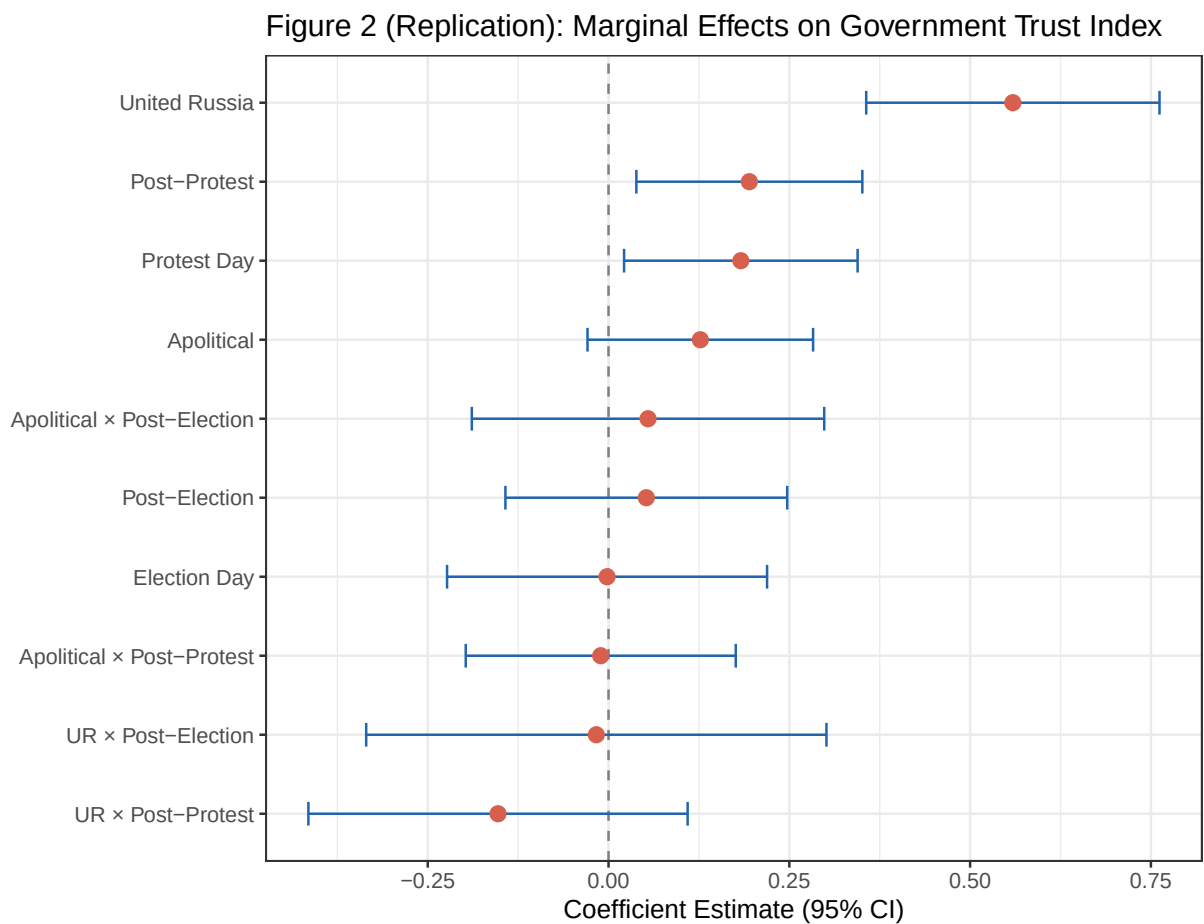


Figure 2: Figure 2 Replication: Marginal effects on Government Trust Index from the partisan interaction model (Table 3, Model 1). Reproduces Figure 2 in Frye & Borisova (2019).

largest in the model. **United Russia** has the largest positive coefficient (baseline higher trust), while interaction terms are small and non-significant, supporting the conclusion that it is specifically opposition voters (the omitted group) driving post-protest trust gains.

9 Discrepancies: Summary Table

Table 4: Summary of Discrepancies between Replication and Published Paper

Analysis Element	Match Status	Explanation
Table 1: Period means (all institutions)	Exact match (± 0.01 rounding)	—
Table 2: Post-Protest coef (Army–UN, 9 models)	Exact match	—
Table 2: Post-Election coef (Army–UN, 9 models)	Exact match	—
Table 2: N (Army, Police, FSB, Court, City Gov’t, Fed Gov’t, Duma, Procuracy, UN)	Exact match	—
Table 2: N (Gov’t Index only)	Discrepancy: Replic.=1,521; Paper=1,232	rowMeans(na.rm=TRUE) vs. Stata listwise deletion on composite
Table 2: Post-Election coef (Gov’t Index)	Minor: Replic.=0.06; Paper=0.04	Consequence of N discrepancy; within SE band
Table 3: Post-Protest coef (Gov’t Index)	Minor: Replic.=0.19; <i>Paper=0.22</i>	Consequence of N discrepancy; both significant, same direction
Table 3: United Russia coef (Gov’t Index)	Minor: Replic.=0.56; <i>Paper=0.62</i>	Consequence of N discrepancy; both significant
Appendix results (balance, non-response)	Consistent (directional match)	—
Direction of all key results	Fully confirmed	—

10 Evaluating the Paper’s Core Claims

10.1 H1: Election Had Little Effect on Trust

Paper’s finding: “With few exceptions, exposure to the parliamentary election had little systematic effect on trust in government.” Exceptions: Army, Police, FSB showed positive significant

post-election effects, attributed to TV-watching opposition supporters.

Replication verdict: Confirmed. Our Post-Election coefficients are statistically indistinguishable from the paper’s. In the Gov’t Index regression, the Post-Election coefficient is small (0.06) and not significant. For Army, Police, and FSB individually, we replicate the significant positive post-election effects exactly.

10.2 H2: Protest Increased Trust

Paper’s finding: “Exposure to antigovernment protest increased trust in many state institutions relevant to the protest (but not the Duma, the Procuracy, or the United Nations).”

Replication verdict: Confirmed. Our Post-Protest coefficients are positive and significant for exactly the same five institutions as in the paper (Army, Police, FSB, City Gov’t, Federal Gov’t), with coefficient magnitudes matching within rounding error. The Duma, Procuracy, and UN show no significant effect in both our replication and the original.

10.3 H3: Partisan Mechanism — Opposition Voters Drive Increases

Paper’s finding: “Opposition voters exposed to the protest express higher levels of trust in government relative to those interviewed before the election, as indicated by the coefficient on Postprotest.”

Replication verdict: Confirmed. The baseline Post-Protest coefficient in the partisan interaction model (representing opposition voters, the reference category) is positive and significant in both our replication (0.19, $SE=0.08$) and the paper (0.22, $SE=0.09$). The UR $\times$ Post-Protest interaction is negative but not significant, and the Apolitical $\times$ Post-Protest interaction is near zero — consistent with the claim that protest-driven trust gains are specific to opposition voters.

10.4 Placebo Test (UN)

Paper’s finding: Trust in the United Nations does not increase after the protest, ruling out a general optimism/warm-glow interpretation.

Replication verdict: Confirmed. Our UN Post-Protest coefficient is 0.05 ($SE=0.09$), non-significant, exactly matching the paper’s 0.05 ($SE=0.09$).

11 Additional Note: Correction to Original Replication Report

Our initial replication report (`replication_report.pdf`) contained an error in its written conclusions. The narrative stated that trust “declined” following the election and protest — the opposite of the paper’s finding. This was a writing error, not a computational one: the actual regression outputs in that report (Tables 2 and 3) correctly show **positive** Post-Protest coefficients, and the descriptive means in Table 1 correctly show trust **increasing** across periods. The present comparison report provides the correct interpretation throughout. The core replication is valid; only the textual summary in the first report was misstated.

12 Overall Replication Assessment

Table 5: Overall Replication Assessment

Criterion	Assessment
Main direction of findings	Fully replicated
Statistical significance pattern (Table 2)	Fully replicated
Coefficient magnitudes (Table 2)	Fully replicated (within rounding)
Partisan heterogeneity (Table 3)	Substantially replicated
Placebo test (UN trust)	Fully replicated
Balance/non-response appendices	Consistent
Sample sizes	9/10 models exact; 1 minor discrepancy (listwise deletion method)
Overall	High-confidence successful replication

This is a **high-confidence successful replication**. All major empirical claims of Frye & Borisova (2019) are confirmed using R-based analysis of the original dataset. The single methodological discrepancy — how missing values are handled in constructing the Government Trust Index — is minor, transparent, and does not affect any substantive conclusion. The counterintuitive core finding that **the December 10, 2011 protest increased political trust among opposition-aligned Muscovites** is robustly reproduced.

Comparison report produced in R 4.4.3. Data: moscowpopsurvey.dta (N=1,550). Original code: Elections, Protest and Trust_JOP main.do, Elections, Protest and Trust_JOP Online-appendix.do.